Levels of Measurement in Quantitative Research

Ordinal data is a type of categorical data where the values have a clear, ordered ranking, but the intervals between the values are not necessarily equal or known.

Examples of ordinal data:

- Survey responses (e.g., *strongly disagree*, *disagree*, *neutral*, *agree*, *strongly agree*)
- Education levels (e.g., *high school*, *bachelor's*, *master's*, *PhD*)
- Pain scales (e.g., *mild*, *moderate*, *severe*)

Key characteristics:

- **Order matters** (there is a ranking)
- **Differences between categories are not measurable or consistent**

This distinguishes ordinal data from:

- **Nominal data** (which has no order, e.g., blood type or gender)
- **Interval/ratio data** (which has measurable and equal intervals, like temperature or height)

The difference between interval and ratio data lies in the **presence of a true zero** and the ability to make **multiplicative comparisons**.

Interval Data

- Numeric data with equal intervals between values, but **no true zero point** (zero is arbitrary, not an absence of the quantity)

Example: Temperature in Celsius or Fahrenheit (0°C ≠ no temperature)

Ratio Data

- **Definition**: Numeric data with equal intervals and a **true, meaningful zero point** (zero = absence of the quantity)

Example: Height, weight, age, income, distance, number of children